

Displaying Describing Summarising Data

Name:

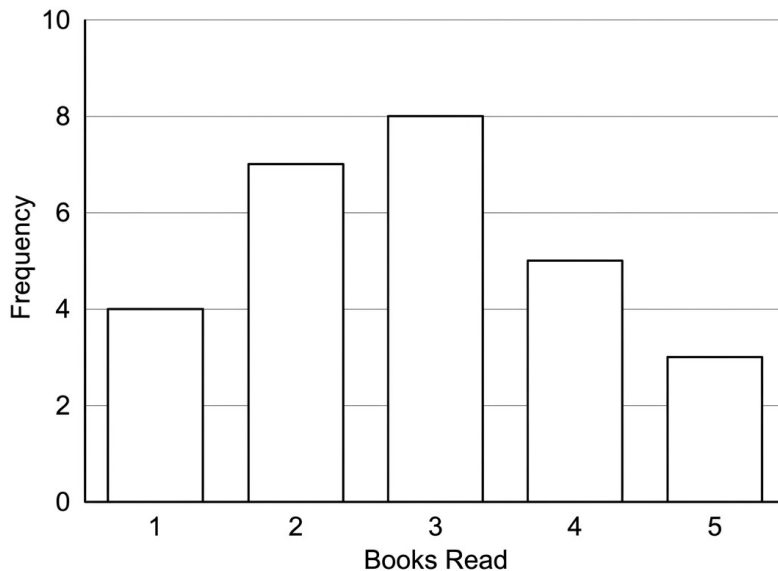
Date:

mathNexa

Success Criteria:

- Interpret frequency charts, dot plots, and histograms
- Calculate mean, median, mode, and range
- Determine interquartile range and identify outliers

1. Look at the frequency bar chart showing the number of books read by students in a month. How many students read 4 or more books?



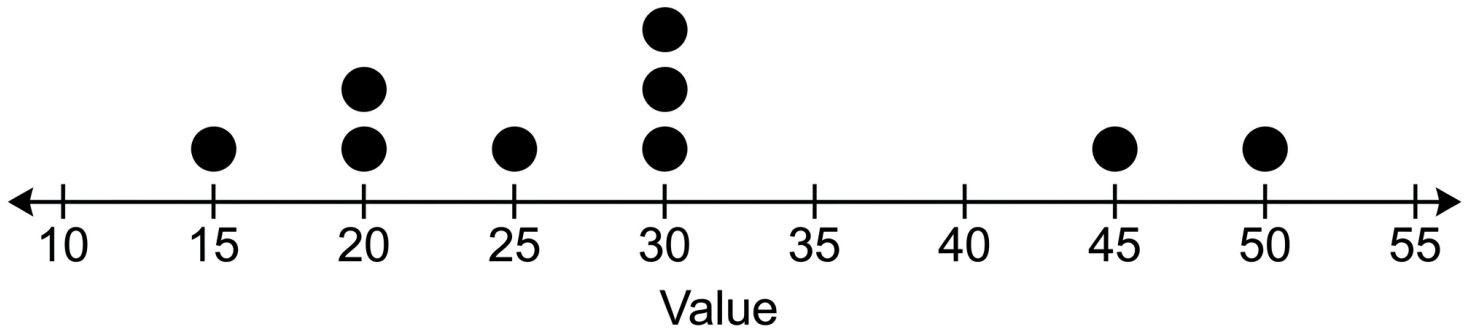
A 16

B 3

C 5

D 8

2. A dot plot shows the number of minutes students spent exercising each day: 15, 20, 20, 25, 30, 30, 30, 45, 50 . What is the mode and the range of this dataset?



Answer: _____

3. A stem-and-leaf plot displays test scores with Key $5 \mid 2 = 52$. If there are 15 data values, what is the median score?

Key: $5 \mid 2 = 52$

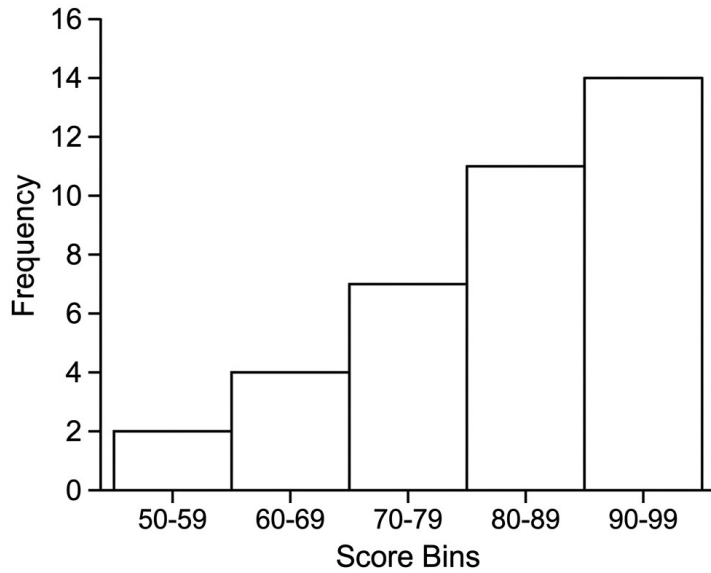
5		2	4	8	
6		1	3	5	9
7		0	4	6	8
8		2	5	7	
9		1			

- A 70 B 74 C 69 D 68

4. Five runners recorded the following times in seconds for a 100 -meter dash: 14.2 , 13.8 , 15.0 , 14.5 , and 13.5 . What is the mean time in seconds?

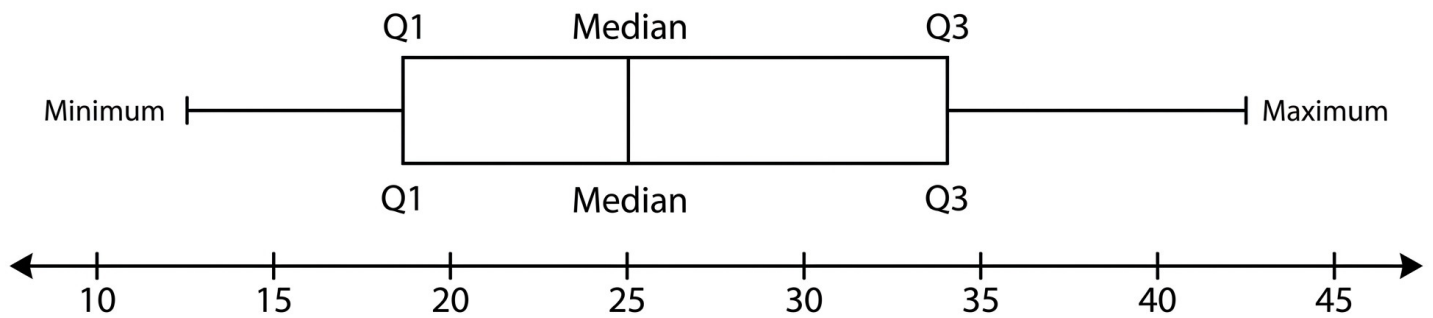
Answer: Mean = _____ seconds

5. Examine the histogram showing test score intervals. Which term best describes the shape of the data distribution?



- ☐ A Uniform
 ☐ B Symmetric
 ☐ C Skewed right
 ☐ D Skewed left

6. A box plot shows a lower quartile (Q_1) of 18 and an upper quartile (Q_3) of 34 for daily temperatures. What is the interquartile range (IQR)?



Answer: IQR = _____

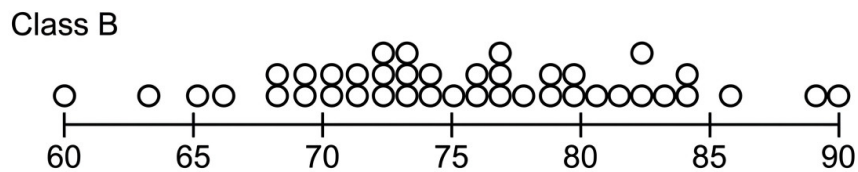
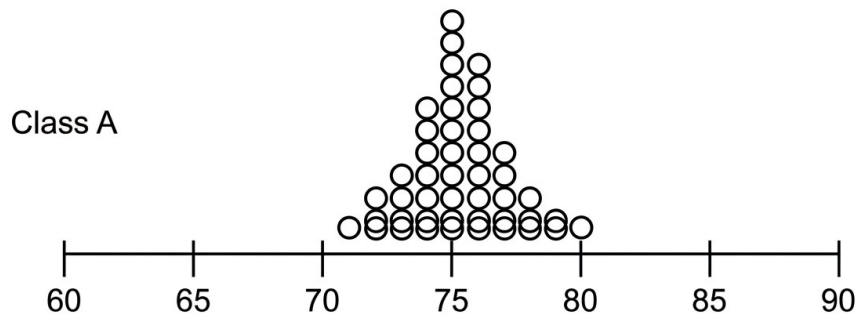
7. A dataset of house prices has one extreme luxury mansion valued at 10 times the other homes. Which measure of central tendency gives the most typical representation of the data?

- ☐ A Median
 ☐ B Mean
 ☐ C Mode
 ☐ D Range

8. The mean score of 4 quiz results is 82. If a student scores 92 on their 5th quiz, what is the new mean score?

Answer: New mean = _____

9. Comparing the dot plots of Class A and Class B test scores where both have a median of 75, which statement accurately compares the variability of the two classes?



- ☐ A Both classes have equal variability
- ☐ B Class B has greater variability
- ☐ C Neither class has any variability
- ☐ D Class A has greater variability
10. A dataset has $Q_1 = 22$ and $Q_3 = 38$. Using the $1.5 \times \text{IQR}$ rule, what is the minimum value a high score must exceed to be considered an outlier?

Answer: Threshold = _____

Answers

1. D

4. 14.2

7. A

2. Mode: 30 , Range: 35

5. D

8. 84

3. A

6. 16

9. B

10. 62